



Degradation of the mechanical properties of root–soil composites under moisture influence

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Abstract

Although the root can enhance the soil's strength, vegetation cover landslide still occurs frequently under the rainfall. To elucidate the mechanism underlying the degradation of the shear strength of root–soil composites under the influence of moisture, we investigated trees from hilly slopes in southeastern China. The tensile mechanical properties of roots were tested under varying moisture conditions. The results of previous work on the friction characteristics of the root–soil interface under different soil water content were also considered. Furthermore, large-scale direct shear tests were performed to assess the strength characteristics of root–soil composites under different root cross-sectional area ratios (RAR) and moisture contents. Based on the widely used Wu model, and incorporating the failure modes of roots in root–soil composites and the mechanism of root–soil interface friction, a root–soil composite strength degradation model was established considering the effects of moisture. Moisture significantly affected the tensile strength of fine tree roots, with the tensile strength of fine roots being lower in the saturated state than in the fresh state. In contrast, coarse roots were almost unaffected by moisture. As the moisture content increased, the additional strength provided by the roots decreased, and the root efficiency (RE_p) decreased significantly. The model was validated against experimental data, and the calculated results were accurate. In root–soil composites, as moisture infiltrates, the tensile strength of the roots, soil shear strength, and root–soil interface shear strength decrease to different degrees. This results in reduced resistance to deformation in the root–soil composites, leading to a decrease in its strength.

Keywords Root reinforcement · Moisture content · Strength degradation model · Root–soil interface friction · Root tensile strength

Introduction

Landslides are common worldwide due to the effects of rainfall, tectonic activities, and human activities. The frequency of landslide disasters has increased about tenfold over the past 50 years (Cendrero et al. 2020; Ozturk et al. 2022). Vegetation plays a crucial role in maintaining the stability of slopes and controlling erosion from mechanical and hydrological perspectives (Veylon et al. 2015; McGuire et al. 2016; Shou and Chen 2021; Xu et al. 2024). Plants can regulate soil moisture content through transpiration, increase

soil matric suction, and reinforce the soil through the roots (Meijer et al. 2016; Liang et al. 2017; Ning et al. 2023).

For slopes with dense vegetation, the combination of high-intensity plant root systems and low-intensity soil forms a root–soil composite (It is referred to as R-SC in this paper), which enhances the shear strength and deformation resistance of the soil through physical and mechanical interactions (Lofalian et al. 2019). Shear strength is a key indicator of the mechanical properties of R-SC, as it partially reflects their resistance to erosion (Ye et al. 2017). Therefore, comprehending soil shear strength is crucial for forecasting soil erosion resistance and slope stability (Jiang et al. 2022). Researchers have assessed the strength characteristics of R-SC under various conditions, including different root contents, configurations, spatial orientations, and soil moisture levels, by conducting indoor direct shear and triaxial tests (Liang et al. 2017; Yildiz et al. 2018; Lian et al. 2019; Forster et al. 2022).

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Numerous researchers have investigated root-soil mechanical interactions and created models to assess the shear strength of R-SC. For instance, Wu et al. (1979a, b) developed a calculation model grounded in the Mohr–Coulomb theory to determine the maximum contribution of roots to soil shear strength, referred to as Wu's model. Its simple parameters and structure have made Wu's model a popular tool for evaluating the reinforcing effect of plant roots under different conditions. Despite its ease of use and broad applicability, Wu's model assumes that all root systems' tensile properties are fully utilized and that roots break at their maximum tensile strength. This assumption tends to overestimate the contribution of roots to soil shear strength (Liang et al. 2020). To better simulate the progressive failure of roots, Pollen and Simon (2005) introduced the Fiber Bundle Model (FBM) algorithm from materials science into the mechanical analysis of root-soil composites. However, the FBM assumes that all the stress from broken roots is transferred to the remaining unbroken roots, neglecting important root-soil interactions. Several root reinforcement models based on the improved FBM algorithm were subsequently proposed, such as the FBM with displacement-controlled loading (Schwarz et al. 2010), the FBM considering the influence of root architecture (Thomas and Pollen-Bankhead 2010), the FBM considering the probability distribution of root failure (Meijer 2021), and the energy-based FBM (Ji et al. 2020). Besides the FBM, other models based on different assumptions have also been proposed. For example, Meijer et al. (2022) considering elasto-plastic behavior of root, root orientation, root failure mode, and the thickness of the shear zone, developed an analytical model to predict mechanical root reinforcement as a function of direct shear displacements. Ning et al. (2023) considered factors such as root–soil interface (It is referred to as R–SI in this paper) friction, root length and constructed an interface bond-based model to calculate the shear strength of R-SC.

Although vegetation can enhance soil strength through mechanical reinforcement, several studies have shown that some slopes with high vegetation coverage frequently experience landslides (Qin et al. 2022), which often occur during heavy rainfall. Rainfall is a major trigger for landslides (Huang et al. 2022a, b). Water infiltration can degrade the physical and mechanical properties of slope soil, leading to increased sliding forces and reduced slope stability (Cuomo and Della 2013). Researchers have extensively studied how moisture weakens soil strength, often explaining this effect through matric suction (Van Genuchten 1980; Fredlund et al. 1994; Nam et al. 2011). While some studies have examined the shear strength of root-soil composites at various soil moisture levels (Huang et al. 2022a, b; Zhu et al. 2022), there is a lack of systematic research on how shear strength changes from the soil's natural moisture content to its saturated state. The mechanism by which moisture affects the shear

strength of R-SC remains unclear. Furthermore, current mechanical models used to calculate the shear strength of R-SC do not account for soil moisture content. There is a need for further research to elucidate the deterioration mechanism and to develop computational models that incorporate the effects of moisture on R-SC strength.

In this study, we investigated shallow landslides with vegetation cover in southeastern coastal China. We collected roots from typical tree species (*Pinus massoniana*, *Cunninghamia lanceolata*) in the region, and tested their tensile strength under various moisture contents. Additionally, we examined the friction characteristics of the root-soil interface under different soil moisture conditions. Large-scale direct shear tests were conducted to analyze the strength characteristics of R-SC at varying moisture contents, taking into account factors such as the impact of moisture on root-soil interface friction and soil matrix suction. An R-SC strength estimation model considering the effect of moisture was established, and the degradation mechanism of the R-SC strength under the influence of moisture was investigated.

Study area

This study was conducted in Wuping County, Longyan City, located in the southwestern region of Fujian Province, China (24°47'–25°29' N; 115°51'–116°23' E; see Fig. 1). On May 26–27, 2022, Wuping County experienced extreme rainfall, with some areas receiving over 200 mm in 24 h, leading to numerous landslides. Shifang Town was the most affected, resulting in eight deaths and the relocation of 1174 people. The landslides in this area were shallow and widespread, with 272 occurrences, the thickness ranged from 0.8 to 2.3 m, and vegetation was carried down along with the soil. On June 16, 2024, the area experienced extreme rainfall again, with the maximum rainfall reaching 372.4 mm in 24 h, starting at 14:00 h in local areas. In total, 47,800 people across 17 townships were affected in the county, with direct economic losses amounting to 415 million yuan. Among the affected areas, the Xiaba, Zhongchi, Xiangdong, and Zhongbao townships experienced multiple landslides, resulting in four deaths and two missing people.

This study focused on a group of landslides located in Shifang Town, Wuping County. The area is predominantly mountainous, with slope surfaces mainly composed of: 1. residual sandy clay soil, 2. completely weathered granite, 3. sand-like highly weathered granite. The surface soil is primarily colluvial sandy clay, physical and mechanical properties of the soil see Table 1 (Lin et al. 2024a). The area has a vegetation cover of 90%, which consists mainly of *C. lanceolata* and *P. massoniana*.

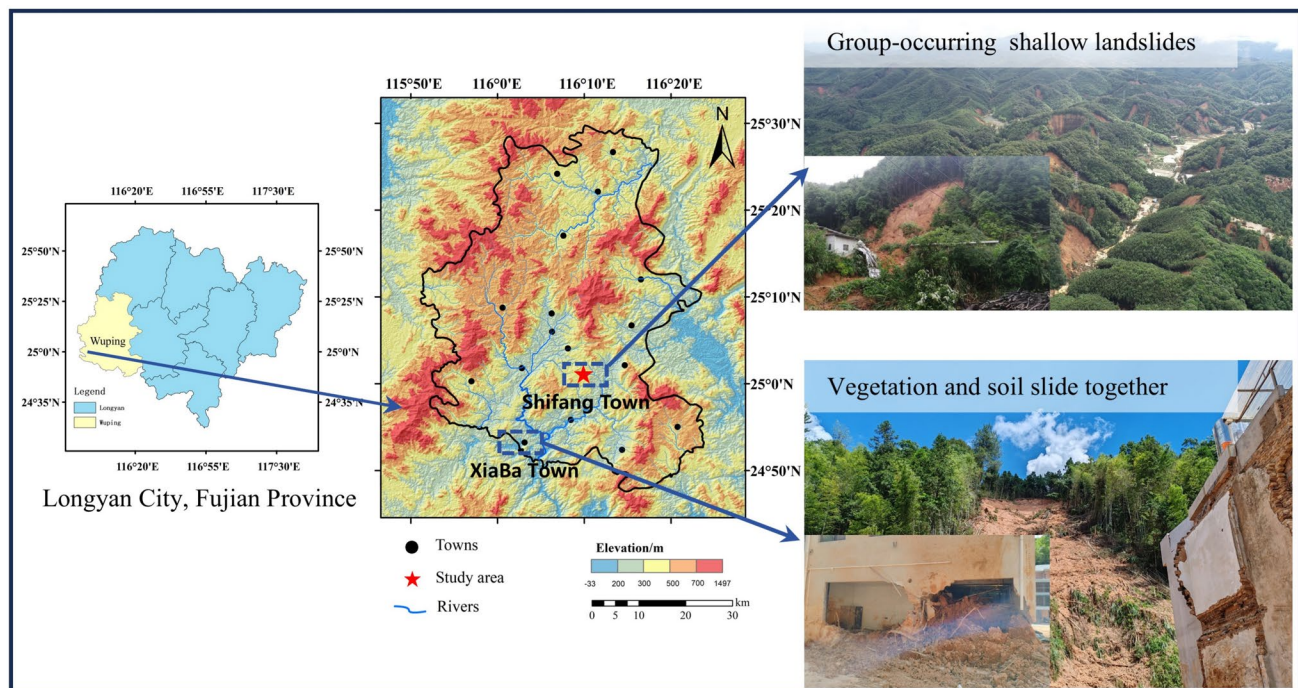


Fig. 1 Study area

Table 1 Physical properties of residual sandy clay soil

Density / g.cm ⁻³	Dry density /g.cm ⁻³	Specific gravity	Sand and gravel content/%
1.63	1.38	2.63	91.2

Material and methods

Tree root tensile tests with different moisture contents

In the study area, *P. massoniana* and *C. lanceolata* of similar ages and grown under similar conditions were selected. Roots of different diameters (see Table 2) were selected from the soil. The roots were refrigerated at 4 °C (Su et al. 2022; Lin et al. 2024b). For both kinds of trees, approximately 823 roots were collected from each species and divided into two groups for processing (Fig. 2). The first group of roots was soaked in water until saturated. The roots were placed in water and weighed every hour. When no additional increase in weight was observed for two consecutive hours, the roots were

considered to be saturated. The mechanical properties of the saturated roots were immediately tested (Zhang et al. 2019). The second group of roots was kept fresh in sealed plastic bags and refrigerated at 4 °C. The mechanical properties were tested within seven days of sampling (For the second group and root tensile tests, see Lin et al. (2024a)). Tensile strength was calculated according to Eq. (1).

$$T = \frac{4F}{\pi D^2} \quad (1)$$

where, T is root tensile strength (MPa); F is root tensile force (N); D is average root diameter (mm).

Direct shear friction test at the root-soil interface with different moisture contents

Sample preparation involved stripping the root bark for specimen creation. A blade was used to peel off the bark, and the bark of roots of the same diameter was fixed onto custom-made acrylic round test blocks with wood adhesive, with the top surface of test block serving as the root surface (Fig. 2). The upper box contained soil with controlled moisture content (with volumetric moisture content adjusted to 25%–45%, and

Table 2 Diameter classes of roots

diameter classes	1	2	3	4	5	6	7
Diameter/(mm)	0.5–1.5	1.5–2.5	2.5–3.5	3.5–4.5	4.5–5.5	5.5–6.5	6.5–7.5

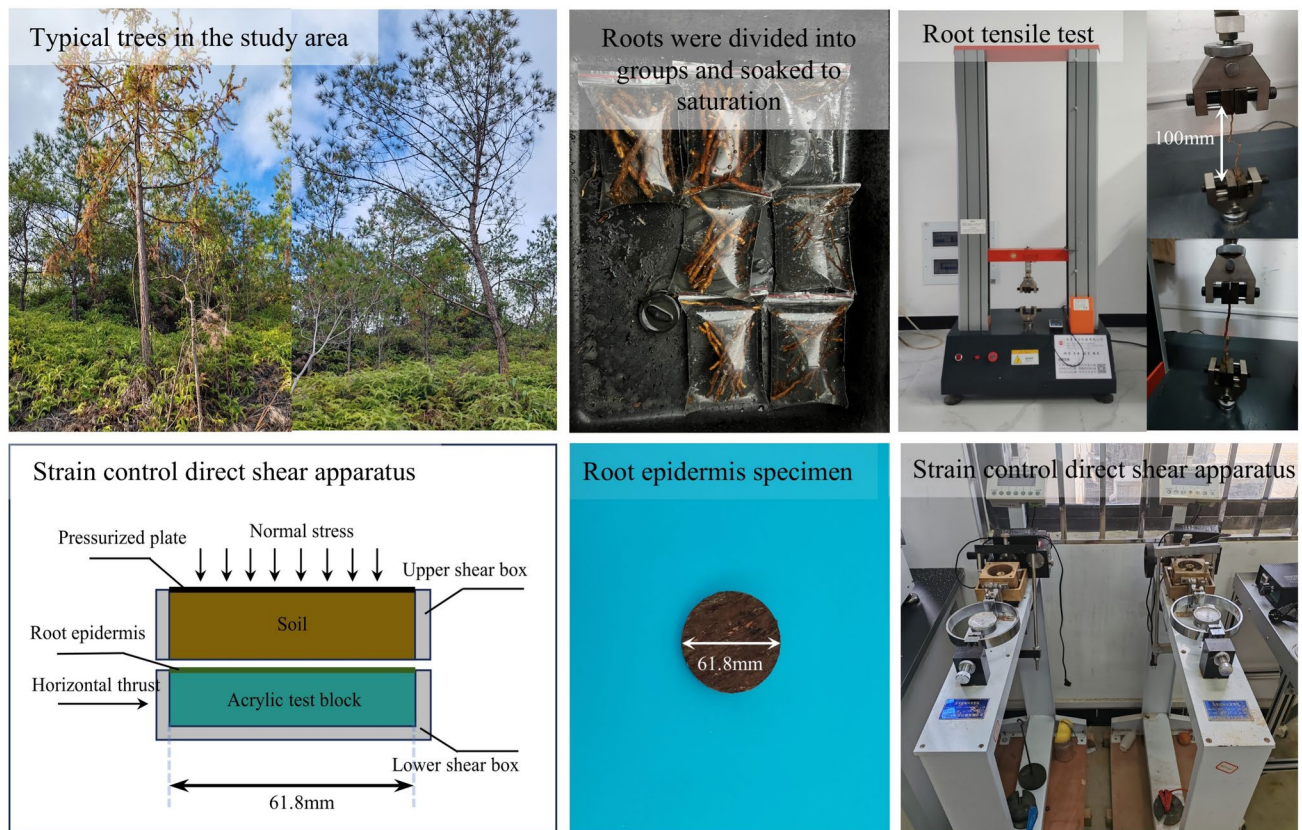


Fig. 2 Tensile testing of roots and direct shear friction testing

the soil was collected from the study area, Table 1). For more detailed experimental procedures, see Lin et al. (2024a).

Large-scale direct shear tests on root–soil composites and SWCC curve measurements of residual sandy clayey soils

This study involved conducting large-scale direct shear tests on tree root–soil composites using a DZJ-300 direct shear apparatus (Fig. 3) (dimensions of the shear box: 15 cm × 15 cm × 12 cm; horizontal displacement: 0–100 mm, accuracy: 0.01 mm; vertical displacement: 0–100 mm, accuracy: 0.01 mm) under different root cross-sectional area ratio (RAR), root diameters (coarse roots, non-coarse roots), root types (*P. massoniana*, *C. lanceolata*), and soil moisture contents. As the shear box height was 12 cm, a root length of 11 cm was selected for the tests. We referred to the studies by Mulia and Dupraz (2006) and Coleman (2007), who classified roots: coarse roots ($D \geq 5$ mm), non-coarse roots ($D: 0–5$ mm), fine roots ($D \leq 2$ mm). The test soil was selected as residual sandy clay from the study area. Since the sliding surface depth in the study area was about 2 m, the vertical pressures selected for this test were 25 kPa,

50 kPa, 75 kPa, 100 kPa (Tan et al. 2019), the shear rate for this test were 0.1 mm/s. The specific test plan is shown in Table 3. Among them, SW25CR8A9–SW45CR8A9 was the control group of R-SC with different moisture contents, SW25CR2A9–W25CR10A9 was the control group with different RAR values. SW25FR4A9 and SW25CR4A9 were R-SC control groups composed of different root diameters within the same RAR and soil moisture content, whereas, SW25CR10A9 and MW25CR10A9 were root–composite control groups composed of roots of different tree species within the same RAR and soil moisture content. TW25–TW45 was the control group of soil without roots at different moisture contents. After obtaining the cohesion and internal friction angle of the R-SC through direct shear tests, we also evaluated the root efficiency of the R-SC.

Root efficiency (RE_p), derived from peak shear strength, evaluates the reinforcement effectiveness of roots within an R-SC (Fan and Su 2008). It is defined as:

$$RE_p = \frac{s^* - s_s}{s_s} = \frac{s_r}{s_s} \quad (2)$$

In the equation, s^* indicates the shear strength of the R-SC (kPa), s_s indicates the shear strength of the soil

Table 3 Test number and test conditions

Sample number	Soil volumetric soil moisture content/(%)	Root diameter /(mm)	RAR/(%)	Angle between root and shear plane /(°)	Type of root
SW25CR6A9	25	5–8 (coarse roots)	0.6	90	<i>C. lanceolata</i>
SW30CR6A9	30	5–8 (coarse roots)	0.6	90	<i>C. lanceolata</i>
SW35CR6A9	35	5–8 (coarse roots)	0.6	90	<i>C. lanceolata</i>
SW40CR6A9	40	5–8 (coarse roots)	0.6	90	<i>C. lanceolata</i>
SW45CR6A9	45	5–8 (coarse roots)	0.6	90	<i>C. lanceolata</i>
SW25CR2A9	25	5–8 (coarse roots)	0.2	90	<i>C. lanceolata</i>
SW25CR4A9	25	5–8 (coarse roots)	0.4	90	<i>C. lanceolata</i>
SW25CR6A9	25	5–8 (coarse roots)	0.6	90	<i>C. lanceolata</i>
SW25CR8A9	25	5–8 (coarse roots)	0.8	90	<i>C. lanceolata</i>
SW25CR10A9	25	5–8 (coarse roots)	1.0	90	<i>C. lanceolata</i>
SW25FR4A9	25	2–4 (non-coarse roots)	0.4	90	<i>C. lanceolata</i>
MW25CR10A9	25	5–8 (coarse roots)	1.0	90	<i>P. massoniana</i>
TW25	25	\	\	\	\
TW30	30	\	\	\	\
TW35	35	\	\	\	\
TW40	40	\	\	\	\
TW45	45	\	\	\	\

without roots, s_r indicates the additional shear strength provided by the roots.

In this study, the soil–water characteristic curve (SWCC) was determined for residual sandy clay in the research area using the Geo-experts soil–water characteristic curve pressure plate device produced by GDS in the UK (Lai et al. 2024). The system consists of four components: a pressure plate device, a control panel, a vertical pneumatic loading system, and a water volume measurement system.

Shear strength estimation model of the root–soil composite considering root–soil interface friction mechanics and matric suction under moisture effect (RSFM model).

Wu et al. (1979a, b), drawing on theories from soil mechanics and material mechanics, developed a simple model to estimate the shear strength of R-SC. This model assumes full utilization of the roots' tensile strength within the composite. It is assumed that during soil failure, all roots reach their ultimate tensile strength and break simultaneously. Consequently, the model determines the roots' mechanical reinforcement impact on the soil using their tensile strength (Fig. 4).

The contribution of roots to soil strength is as follows (Wu et al. 1979a, 1979b):

$$s_r = \sigma_r \tan \phi' + \tau_r = t_r (\cos \theta \tan \phi' + \sin \theta) \quad (3)$$

where:

$$t_r = \sum F_{ri} / A \quad (4)$$

Here, t_r indicates the average tensile strength of the roots per unit area of soil (MPa), $\sum F_{ri}$ indicates the sum of the tensile forces of the roots within the shear zone, A indicates the area of the shear zone (cm^2), ϕ' indicates the effective internal friction angle, and θ indicates the shear deformation angle ($^\circ$).

Wu et al. (1979a, b) conducted a sensitivity analysis and found that $(\cos \theta \tan \phi' + \sin \theta)$ is not sensitive to the value of θ . When $48^\circ < \theta < 72^\circ$, $(\cos \theta \tan \phi' + \sin \theta)$ is close to 1.2. Thus, Eq. (3) can be simplified to:

$$s_r = k t_r \quad (5)$$

The value of k is generally considered to be 1.2, and because $t_r = T_r (A_s / A)$, where A_s indicates the total root area (cm^2) in the region, T_r is the average tensile strength of root, Eq. (3) can also be expressed as:

$$s_r = 1.2 T_r (A_s / A) \quad (6)$$

According to Eq. (6), in the Wu model, the enhancement in soil shear strength due to roots is influenced by the RAR and T_r . When roots are oriented perpendicularly to the shear plane, the shear strength of the R-SC can be derived using the Mohr–Coulomb equation (Wu et al. 1979a, 1979b; Ning et al. 2023):

$$s^* = s_s + s_r = \sigma \tan \varphi + c + s_r \quad (7)$$

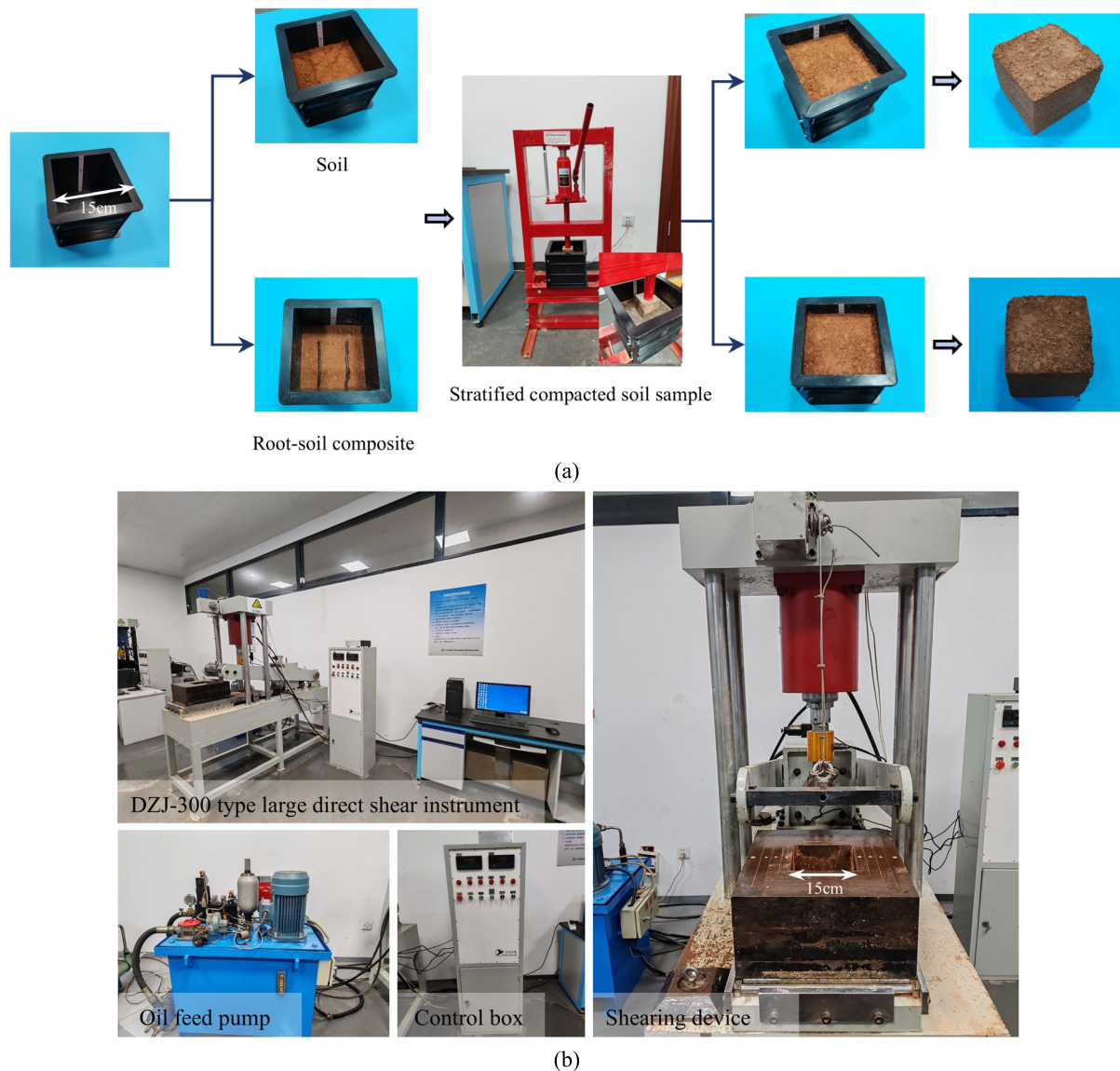


Fig. 3 Large-scale direct shear tests. (a) Production process of the soil samples. (b) DZJ-300 large-scale direct shear apparatus

In the equation, σ indicates the normal stress on the shear plane (kPa), c indicates cohesion of the soil (kPa), and φ indicates the internal friction angle of the soil ($^{\circ}$).

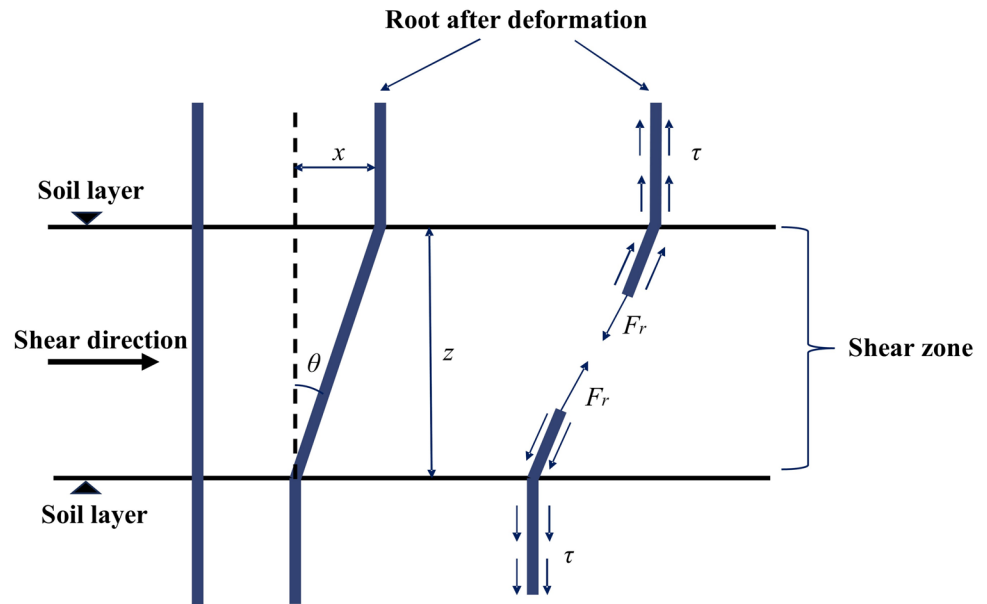
However, in practice, the tensile strength of R-SC roots cannot be fully utilized (Liang et al. 2020; Fan and Tsai 2016). When the bonding strength at the root-soil interface (Some scholars also call it the shear strength of the root-soil interface (Gan et al. 2023)) is low, roots may slip out before breaking. In practice, root failure in the R-SC frequently involves both partial breakage and slipping. Therefore, when calculating the s^* , the shear strength at the R-SI and tensile strength of roots need to be comprehensively considered. When the R-SC experiences shear

stress, the bonding effect at the R-SI transforms the soil's shear stress into tensile stress on the roots. Ning et al. (2023) expanded the Wu model by including this bonding effect, defining the bonding strength (shear strength) as the shear strength at the root-soil contact surface during shearing. This strength encompasses both interfacial cohesion and friction, consistent with the Mohr–Coulomb law (Xia et al. 2021; Sukmak et al. 2020).

$$\tau_{rs} = P_r f_{rs} + C_{rs} = P_r \tan \varphi_{rs} + C_{rs} \quad (8)$$

In the equation, τ_{rs} indicates the R-SI shear strength, C_{rs} indicates the R-SI cohesion (kPa), f_{rs} indicates the R-SI friction coefficient, P_r indicates the normal stress at

Fig. 4 Stress conditions of roots in the R-SC



the interface (kPa), and φ_{rs} indicates the internal friction angle of the interface ($^{\circ}$).

The root-soil interface cohesion (C_{rs}) is generally composed of interface soil cohesion and interface chemical adhesion. The interaction between soil particles and the root surface forms the interface soil cohesion, while the chemical substances secreted from the root surface during growth, such as macromolecular polysaccharides, organic acids, and sugar alcohols, form the interface chemical adhesion (the remolded root-soil composite does not contain interface chemical adhesion) (Oleghe et al. 2017; Naveed et al. 2018; Xia et al. 2021). When relative sliding occurs locally at the root-soil interface, the root-soil interface cohesion fails (Fan and Su 2008; Ning et al. 2023).

Due to the roughness of the root epidermis, some soil particles embed into the uneven surface of the roots. When the roots are subjected to tension and tend to displace, the local protrusions on the root surface compress the surrounding soil, and the mechanical interlocking between the root-soil interface forms strong interfacial friction resistance (Liu et al. 2018; Xia et al. 2021). The interfacial friction is determined jointly by the interfacial normal stress and the friction coefficient of the interface.

During rainfall, as water infiltrates, the soil volumetric moisture content (θ_w) increases and the R-SI shear strength (bonding strength) gets affected. The shear strength of the R-SI is composed of two parts: the R-SI cohesion and the R-SI friction coefficient. The R-SI shear strength correction factor, denoted as $k_{rs\theta f}$ and $k_{rs\theta c}$, is defined as the ratio of shear strength at the R-SI under varying moisture contents to the shear strength under natural moisture conditions. In this research, the natural soil

volumetric moisture content was about 25%. The expression for the R-SI friction coefficient correction factor ($k_{rs\theta f}$), which considers the relationship between the R-SI friction coefficient (f_{rs}) and volumetric water content (θ_w), is as follows:

$$k_{rs\theta f} = a_f + b_f \theta_w \quad (9)$$

The expression for the R-SI cohesion correction factor ($k_{rs\theta c}$), which considers the relationship between the R-SI cohesion (C_{rs}) and volumetric water content (θ_w), is as follows:

$$k_{rs\theta c} = a_c \theta_w^{b_c} \quad (10)$$

In the equation, a_c , b_c , a_f , and b_f is the interface parameters.

The R-SI shear strength at different moisture contents is as follows:

$$\tau_{rs\theta} = k_{rs\theta f} P_r f_{rs} + k_{rs\theta c} C_{rs} = k_{rs\theta f} P_r \tan \varphi_{rs} + k_{rs\theta c} C_{rs} \quad (11)$$

A force analysis was performed on the root segment below the shear plane in Fig. 5 (as shown in Fig. 6) to obtain the following force equilibrium equation (Xie 1990; Ning et al. 2023):

$$\int_0^{T_r} dT_r = \int_0^L \tau_{rs} \pi D_r dl \Rightarrow F_r = \tau_{rs} \pi D_r L \quad (12)$$

Here, F_r indicates the tensile force of the root (kN), D_r indicates the diameter of the root (m), and L indicates the length of the root below shear plane (m).

Therefore, the tensile stress of the root can be expressed as (Xia et al. 2021):

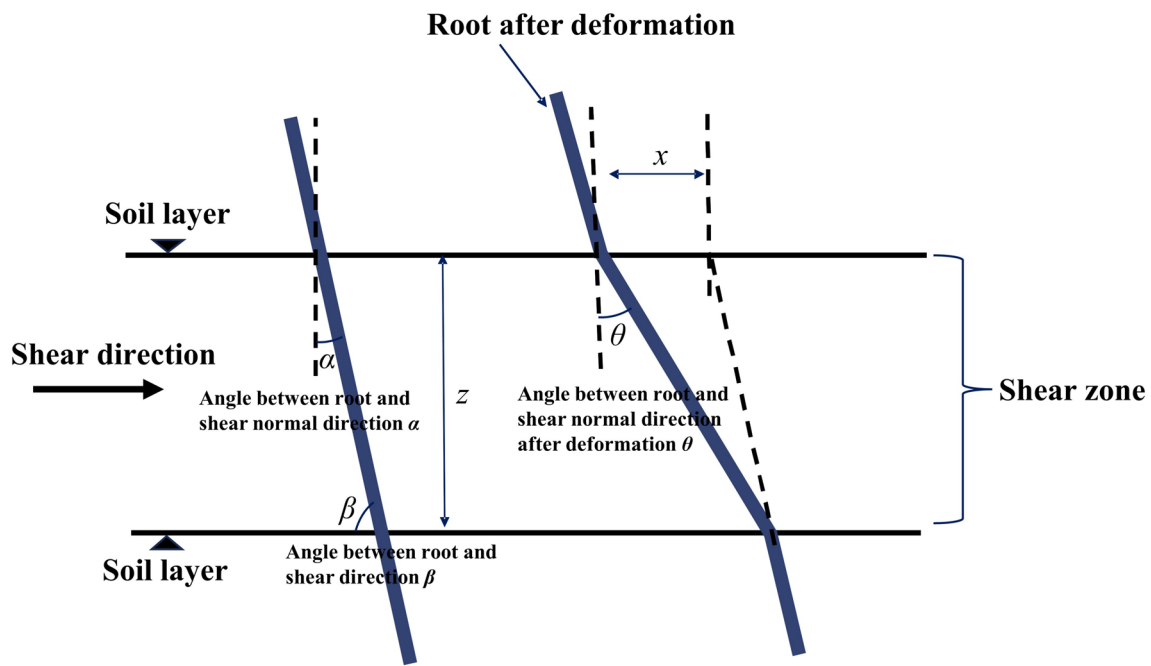
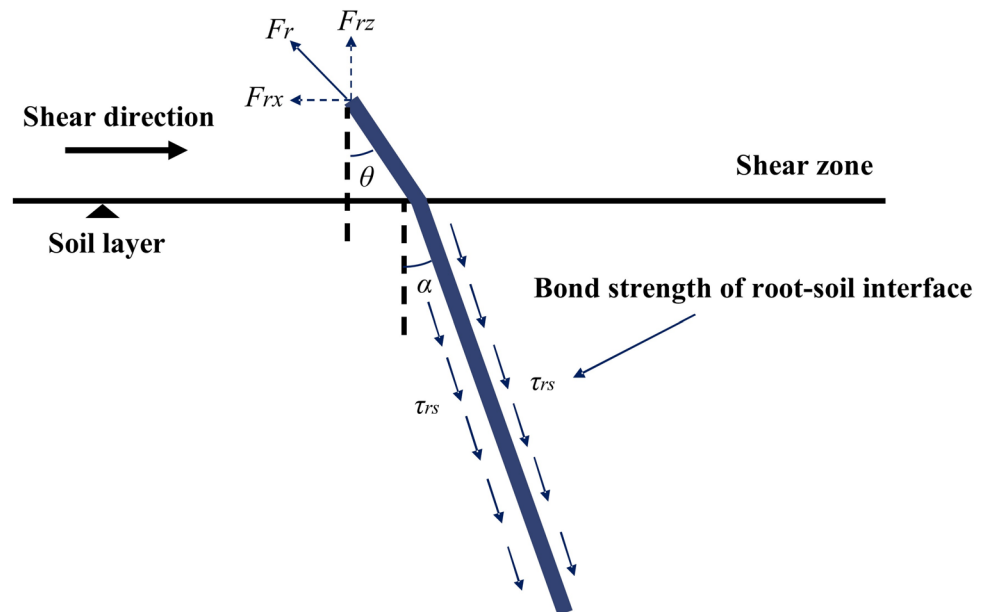


Fig. 5 Deformation of roots in the R-SC

Fig. 6 Stress analysis of the root segment



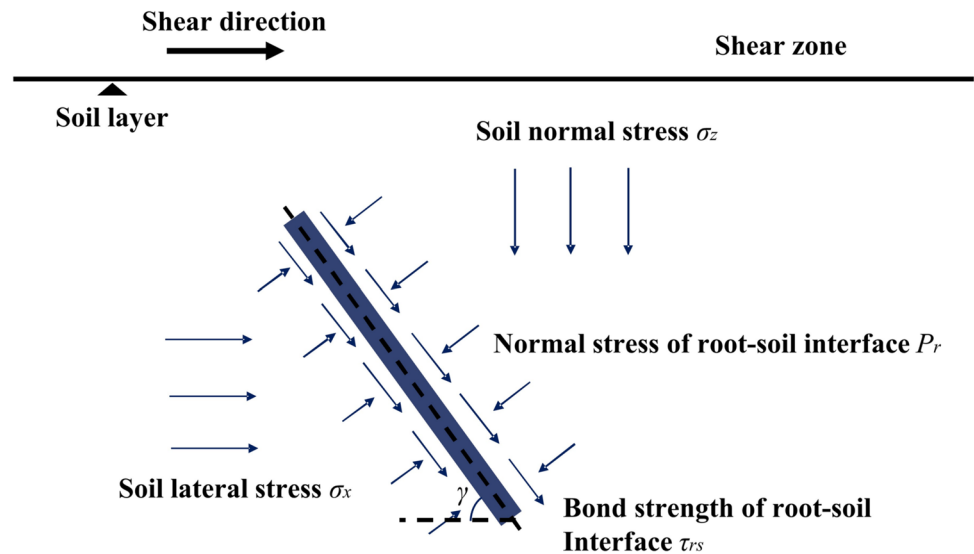
$$t_{rs} = \frac{4F_r}{\pi D_r^2} = \frac{4\tau_{rs}L}{D_r} \quad (13)$$

Assuming that the shear direction aligned with the soil's lateral stress direction, a force analysis was conducted on a segment of the root below the shear plane, as illustrated in Fig. 7. Let the root system length be L . Under force equilibrium conditions, the angle between the root and the lateral stress direction of the soil is γ , and

the angle between the root and the shear direction is β . Assuming that the shear direction is consistent with the lateral stress direction of the soil, $\gamma = \beta$. When $\gamma < 90^\circ$, the normal stress on the R-SI of the inclined root system is (Xia et al. 2021):

$$P_r L = \sigma_x \sin \gamma \cdot L \sin \gamma + \sigma_z \cos \gamma \cdot L \cos \gamma \\ \Rightarrow P_r = \sigma_x^2 \sin^2 \gamma + \sigma_z^2 \cos^2 \gamma \quad (14)$$

Fig. 7 Analysis of root forces in soil



When $\gamma = 90^\circ$, the root is perpendicular to the shear direction, and the normal stress on the root-soil interface is (Xia et al. 2021):

$$P_r L = \sigma_x L \Rightarrow P_r = \sigma_x = k_0 \sigma_z \quad (15)$$

In the equation, k_0 represents the static lateral pressure coefficient of the soil, which varies with different soil types. It can be determined through static lateral pressure tests of soil or obtained via empirical equations: $k_0 = 1 - \sin \varphi$ (Xia et al. 2021) or reference manuals.

When an R-SC undergoes deformation due to shear, the root system fails in one of two modes: breakage or pullout (Fan 2012; Ning et al. 2023). In root-soil composites, roots of various diameters typically coexist. During the failure process of the root-soil composite, some roots may fail due to tensile breakage, while others may fail due to pullout. In this case, the additional strength provided by all roots is generally the sum of the cohesion contributed by roots undergoing tensile breakage and those undergoing pullout failure. Before calculation, the failure mode of each root must be determined, and the actual additional strength provided is often governed by the weaker of the two failure modes. Therefore, the additional strength contributed by the root system in the R-SC is the sum of the additional strength contributions of these two failure modes:

$$s_r = s_{rl} + s_{rd} \quad (16)$$

In the equation, s_{rl} represents the additional strength provided by the roots that fail in tensile breakage in the R-SC, and s_{rd} represents the additional strength provided by the roots that fail in pullout.

In some cases, all roots in the R-SC may fail due to tensile breakage, in which case $s_r = s_{rl}$; alternatively, all roots may fail due to pullout, in which case $s_r = s_{rd}$.

For each root in the R-SC, if the root embedded in the soil is deep enough that the bonding force (cohesion and friction at the root-soil interface) at the R-SI exceeds the ultimate tensile strength of the root, and if the tensile stress in the root exceeds its ultimate tensile strength during loading, the root will fail in break apart. In this failure mode, the additional strength provided by a single root in the R-SC is equal to the ultimate tensile strength of the root.

$$s_{rl} = \sigma_r \tan \phi' + \tau_r = t_{rl} (\cos \theta \tan \phi' + \sin \theta) = 1.2 T_{rl} (A_{sl}/A) (\cos \theta \tan \phi' + \sin \theta) \quad (17)$$

In the equation, t_{rl} represents the average tensile strength (kPa) produced by roots failing in tensile breakage per unit area of soil, and $t_r = T_{rl} (A_{sl}/A)$, A_{sl} represents the root area of roots with tensile breakage failure in the root-soil composite within the region (cm^2). T_{rl} is the average tensile strength of the roots with tensile breakage failure in the root-soil composite (Mpa). When the root-soil composite is in an unsaturated state, the tensile strength of the roots is selected as the tensile strength measured in the fresh state; when the root-soil composite is in a saturated state, the tensile strength of the roots is selected as the tensile strength measured in the saturated state.

If the root embedded depth below the shear plane is shallow, making the adhesion force at the R-SI less than the root's ultimate tensile strength, and during loading the root's tensile stress is less than its ultimate tensile strength, the root will fail by pullout. In this failure mode, the additional strength provided by a single root in the R-SC is supplied by the bonding force at the R-SI.

$$s_{rd} = \sigma_r \tan \phi' + \tau_r = t_{rs} (A_{sd}/A) (\cos \theta \tan \phi' + \sin \theta) \Rightarrow s_{rd} = \frac{4(P_r \tan \phi_{rs} + C_{rs})L}{D_r} (A_{sd}/A) (\cos \theta \tan \phi' + \sin \theta) \quad (18)$$

In the equation, A_{sd} represents the root area of roots with pullout failure in the root-soil composite within the region (cm^2).

The angle between the deformed root and the shear normal direction can be determined by shear displacement (Gray and Alrefeai 1986):

$$90^\circ - \theta = \tan^{-1} \left[\frac{1}{(m + (\tan 90^\circ - \alpha)^{-1})} \right] \quad (19)$$

In the equation, m represents the shear deformation ratio, which can be determined by the following equation:

$$m = x/z \quad (20)$$

In the equation, x represents the shear displacement, and z represents the width of the shear zone (as shown in Fig. 5).

If the shear displacement and shear zone width cannot be determined, the approach proposed by Wu et al. can be used, which suggests that $(\cos \theta \tan \theta' + \sin \theta)$ is not sensitive to θ and is approximately 1.2 when $48^\circ < \theta < 72^\circ$ (Wu et al. 1979a, b). Therefore, Eq. (18) can be expressed as:

$$s_{rd} = 1.2 \frac{4(P_r \tan \varphi_{rs} + C_{rs})L}{D_r} (A_{sd}/A) \quad (21)$$

During rainfall, as water infiltrates the soil and the moisture level increases, the cohesion at the R-SI changes. Under such conditions, the additional strength provided by the roots in the R-SC can be expressed as:

$$s_{rd\theta} = 1.2 k_{r\theta} t_{rs} (A_{sd}/A) = 1.2 \frac{4(k_{rs\theta} P_r \tan \varphi_{rs} + k_{rs\theta} C_{rs})L}{D_r} (A_{sd}/A) \quad (22)$$

In the equation, $s_{rd\theta}$ represents the additional strength provided by the roots that fail in a pullout considering the effect of moisture.

In an in situ direct shear test, for the shear plane of the R-SC, the normal stress is as follows:

$$\sigma_z = \rho Vg/A = (\rho_d + \rho_d w) Vg/A = (\rho_d + \rho_d w) gh \quad (23)$$

As many soil moisture sensors used in landslide monitoring operate on the principles of time domain reflectometry (TDR), the measured moisture content is the volumetric moisture content. The conversion relationship between the volumetric moisture content and gravimetric moisture content is (Lin et al. 2024b):

$$\theta_w = V_w/V = w\rho_d/\rho_w \quad (24)$$

Here, θ_w represents the volumetric moisture content (%), w denotes the gravimetric moisture content (%), V_w represents the volume of pore water (cm^3), V represents the total volume of the soil sample (cm^3), ρ_d represents the dry density of the

soil (g/cm^3), and ρ_w and represent the density of the pore water (g/cm^3). Therefore, Eq. (23) can be expressed as:

$$\sigma_z = (\rho_d + \theta_w \rho_w) Vg/A = (\rho_d + \theta_w \rho_w) gh \quad (25)$$

For vegetation-covered landslides, the soil is often unsaturated. When rainfall infiltrates, the shear strength of unsaturated soil can vary significantly with changes in moisture conditions. In unsaturated soil, suction is commonly used to describe the moisture state of the soil. Many researchers have proposed various equations for the shear strength of unsaturated soils. Currently, depending on the choice of stress variables, the theories of unsaturated soil strength can be broadly classified into two categories. One category is the extension of the effective stress principle for saturated soils to unsaturated soils, as proposed by Bishop (1959), which introduces a formula for the shear strength of unsaturated soils using effective stress as a single state variable:

$$\tau_f = \tau_s + \tau_{us} = c' + [(\sigma - u_a) + \chi\psi] \tan \varphi' \quad (26)$$

Here, τ_f represents the shear strength of unsaturated soil (kPa), τ_s represents the shear strength of saturated soil (kPa), τ_{us} represents the suction strength (kPa), c' represents the effective cohesion of saturated soil (kPa), φ' represents the effective angle of internal friction ($^\circ$), $\sigma - u_a$ represents the net normal stress on the shear plane at failure (kPa), ψ represents the suction (kPa), and χ represents an unsaturated effective stress parameter, where $\chi = 1$ indicates soil saturation and $\chi = 0$ indicates no moisture content in the soil.

The other category is the linear unsaturated soil strength theory proposed by Fredlund et al. (1978); it uses net stress and suction as dual stress variables.

$$\tau_f = c' + (\sigma - u_a) \tan \varphi' + \psi \tan \varphi^b \quad (27)$$

Here, φ^b represents the internal friction angle corresponding to the matric suction ($^\circ$).

The above equations are established based on the effective stress theory. Among them, the parameter χ in Bishop's formula and the parameter φ^b in Fredlund's formula need to be determined through unsaturated soil triaxial or direct shear tests. However, in many practical engineering applications, controlling the suction and measuring the pore water pressure are relatively difficult. Therefore, some researchers have simplified the unsaturated soil strength models. Based on Bishop's strength formula, Khalili and Khabbaz (1998) fitted a large amount of experimental data and proposed that a functional relationship exists between the unsaturated effective stress parameter and the suction ratio ψ/ψ_a :

$$\tau_f = \tau_s + \tau_{us} = \tau_s + \psi \left(\frac{\psi}{\psi_a} \right)^{-\eta} \tan \varphi' \quad (28)$$

Here, η indicates the fitting parameter and ψ_a indicates the air entry value (kPa).

From Eqs. (26)–(28), it can be seen that the shear strength of unsaturated soil, in addition to being related to c', φ' , and $\sigma - u_a$ like saturated soil, is also related to suction. The relationship between suction strength τ_{us} and suction ψ is the key to calculating the strength of unsaturated soil. Soil moisture content is a key factor affecting the magnitude of matric suction. Scholars refer to the curve expressing the relationship between soil matric suction and water content (saturation) changes as the SWCC. For the residual sandy clay soil, the commonly used SWCC model, such as the Van Genuchten model (Van Genuchten 1980), is expressed as follows:

$$\theta_w = \theta_r + \frac{\theta_s - \theta_r}{(1 + (a\psi)^b)^{\frac{b-1}{b}}} \quad (29)$$

Here, θ_s and θ_r indicate the saturated volumetric moisture content and residual volumetric moisture content, respectively, ψ indicates the suction, and a and b indicate the fitting parameters.

Due to the difficulty in measuring the residual moisture content of residual soils, some researchers have modified the VG model for residual soils without considering the influence of residual moisture content (Xu 2015):

$$\theta_w = \frac{\theta_s}{(1 + (a\psi)^b)^{\frac{b-1}{b}}} \quad (30)$$

By transforming Eq. (30), we get:

$$\psi = \frac{1}{a} \left[\left(\frac{\theta_s}{\theta_w} \right)^{\frac{1}{b-1}} - 1 \right] \quad (31)$$

The shear strength of soil considering moisture content $s_{s\theta}$ is:

$$s_{s\theta} = \tau_s + \tau_{us} = (\rho_d + \theta_w \rho_w) h g \tan \varphi' + c' + \tau_{us} \quad (32)$$

Based on the above, the equation for calculating the shear strength of the R-SC, considering the variation in soil moisture content, is as follows:

$$\Rightarrow s_{\theta}^* = (\rho_d + \theta_w \rho_w) h g \tan \varphi' + c' + \tau_{us} + 4.8 k_{r\theta} L (k_{rs\theta f} P_r \tan \varphi_{rs} + k_{rs\theta c} C_{rs}) (A_{sd}/A)/D_r + 1.2 T_{rl} (A_{sl}/A) \quad (33)$$

In the equation: $s_{r\theta}$ represents the root additional strength considering the effect of moisture in root-soil composite, where $s_{r\theta} = s_{rl} + s_{rd\theta}$. A_{sl} represents the root area of roots with tensile breakage failure in the root-soil composite within the region (cm^2). T_{rl} is the average tensile strength of the roots with tensile breakage failure in the root-soil composite (MPa). A_{sd} represents the root area of roots with pullout

failure in the root-soil composite within the region (cm^2). the relationship between ψ and θ_w can be obtained from Equation (31), while the correction factors $k_{rs\theta f}$ and $k_{rs\theta c}$ in relation to θ_w can be determined from Equations (9) and (10). Before calculation of equation (33), for each root, the failure mode should be determined based on parameters such as the root diameter, root length, and root-soil interface friction characteristics. First, s_{rl} is calculated using equation (16) for the tensile breakage failure mode, and s_{rd} is calculated using equation (17) for the pullout failure mode. Since the actual cohesion provided by the root is usually the weaker of the two failure modes, the smaller value between s_{rl} and s_{rd} should be taken to determine the failure mode of each root. The calculation flow of the RSFM model is shown in Fig. 8.

Results

Influence of moisture on root tensile strength

Under fresh conditions, 70 *P. massoniana* root samples were tested for tensile strength (the root diameter: 0.72–7.47 mm, tensile strength: 7.16–33.48 MPa) (Fig. 9(a)). The Kruskal–Wallis test results indicated significant differences in tensile strength among fresh *P. massoniana* roots of different diameters ($P < 0.01$). (The test results of root tensile strength in the fresh conditions in this paper are from Lin et al. 2024a, 2024b). Under saturated conditions, 70 *P. massoniana* root samples were tested for tensile strength (the root diameter: 0.77–7.43 mm, tensile strength: 5.77–25.89 MPa) (Fig. 9(b)). The Kruskal–Wallis test results indicated significant differences in tensile strength among saturated *P. massoniana* roots of different diameters ($P < 0.01$). The tensile strength of the *P. massoniana* roots (under fresh and saturated conditions) decreased with an increase in diameter, indicating a negative power function correlation. The relationship between the root tensile strength (T) and diameter (D) of *P. massoniana* roots under fresh conditions was $T = 32.46D^{-0.66}$ ($R^2 = 0.86$, $P < 0.01$) (Fig. 9(a)) (Lin et al. 2024a). For *P. massoniana* roots under saturated conditions was $T = 20.32D^{-0.49}$ ($R^2 = 0.83$, $P < 0.01$) (Fig. 9(b)).

Under fresh conditions, 70 *C. lanceolata* root samples were tested for tensile strength (the root diameter: 0.51–7.49 mm, tensile strength: 7.26–60.95 MPa) (9(a)). The Kruskal–Wallis test results indicated significant differences in tensile strength among fresh *C. lanceolata* roots of different diameters ($P < 0.01$) (Lin et al. 2024a, 2024b). Under saturated conditions, 70 *C. lanceolata* root

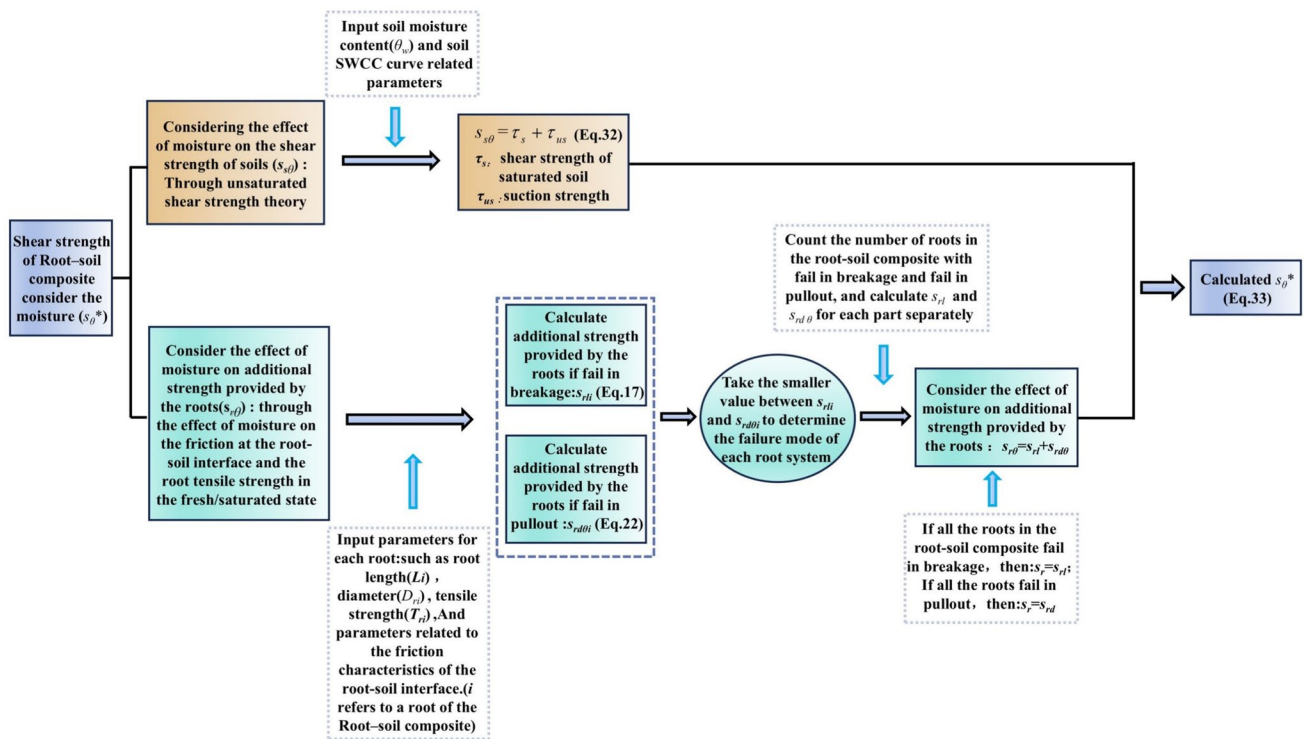


Fig. 8 Calculation flow of the RSFM model

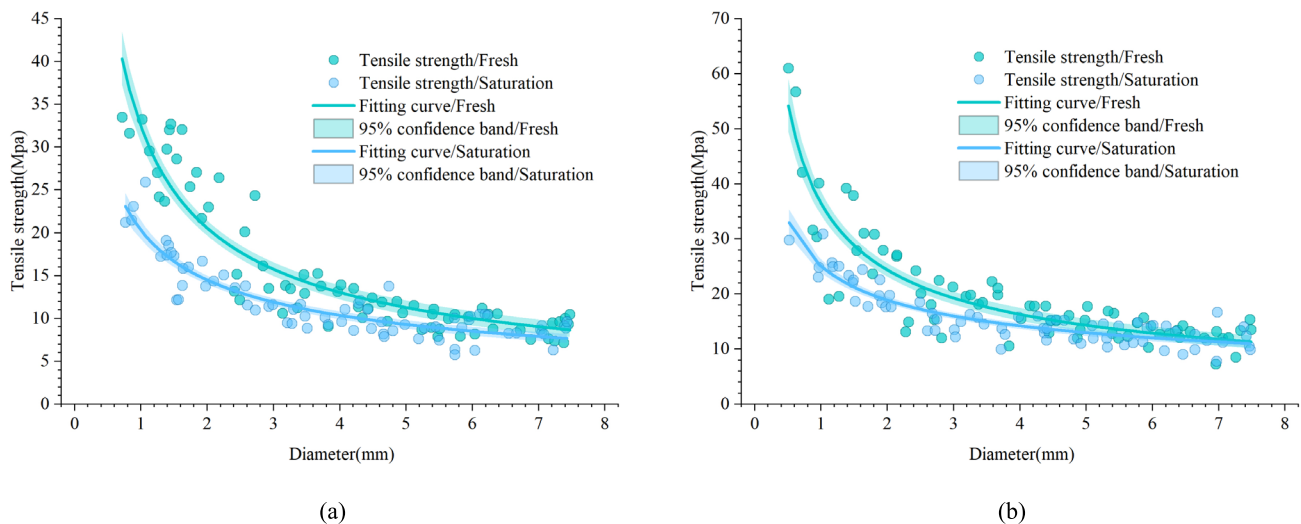


Fig. 9 Tensile strength of the roots in the fresh/saturated state. (a) Tensile strength of *P. massoniana* in fresh/saturated state; (b) Tensile strength of *C. lanceolata* in fresh/saturation state

samples were tested for tensile strength (the root diameter: 0.52 to 7.48 mm, tensile strength: 7.74–30.87 MPa) (Fig. 9(b)). The Kruskal–Wallis test results indicated significant differences in tensile strength among saturated *C. lanceolata* roots of different diameters ($P < 0.01$). The tensile strength of the *C. lanceolata* roots (under

fresh and saturated conditions) decreased as the diameter increased, indicating a negative power function correlation (Fig. 8(b)). The relationship between the T and D of *C. lanceolata* roots under fresh conditions was $T = 36.55D^{-0.58}$ ($R^2 = 0.81$, $P < 0.01$) (Fig. 9(a)) (Lin et al.

2024a). For *C. lanceolata* roots under saturated conditions was $T = 25.12D^{-0.41}$ ($R^2 = 0.82$, $P < 0.01$) (Fig. 9(b)).

The results revealed that tensile strength differed between the fresh and saturated states of the roots. The Mann–Whitney U test results revealed a significant difference in tensile strength of *P. massoniana* roots between the fresh and saturated states ($P < 0.05$). A significant difference was also observed in the tensile strength of *C. lanceolata* roots when comparing the fresh and saturated states ($P < 0.05$). When the roots transitioned from the fresh state to the saturated state, the average tensile strength of the *P. massoniana* roots decreased from 15.47 MPa to 11.69 MPa, indicating a reduction of 24.43%, and the average tensile strength of the *C. lanceolata* roots decreased from 19.72 MPa to 15.67 MPa, indicating a reduction of 20.54%. The violin plot (In the figure, the circles represent the mean value of each dataset, the width of the violin body indicates the data density, and the height of the violin body reflects the range of data distribution for the group.) (Fig. 10) of roots tensile strength in fresh/saturated state across different diameter classes revealed that moisture significantly affected the tensile strength of smaller-diameter roots (diameter classes 1–3) for both two species of trees. The tensile strength of fine roots in the saturated conditions was considerably lower than that in the fresh conditions, whereas the effect on larger-diameter roots (diameter classes 6–7) was almost negligible.

Effect of moisture on friction characteristics of root-soil interface

Figure 11 shows that, within a soil volumetric moisture content (θ_w) range of 25%–45%, the cohesion (C_{rs}) and friction

coefficient (f_{rs}) at the R–SI decreases significantly. For *P. massoniana*, C_{rs} decreases from 11.74 kPa to 8.33 kPa, with the relationship between θ_w and C_{rs} described by $C_{rs} = 15.73 - 0.17\theta_w$ ($R^2 = 0.89$, $P < 0.01$). For *C. lanceolata*, C_{rs} decreases from 17.06 kPa to 13.24 kPa, relationship given by: $C_{rs} = 22.49 - 0.21\theta_w$ ($R^2 = 0.96$, $P < 0.01$). For *P. massoniana*, f_{rs} decreases from 0.63 to 0.57, with the relationship between θ_w and f_{rs} described by: $f_{rs} = 1.19 - 0.19\theta_w$ ($R^2 = 0.97$, $P < 0.01$). For *C. lanceolata*, f_{rs} decreases from 0.53 to 0.43, with the relationship: $f_{rs} = 1.77\theta_w^{-0.37}$ ($R^2 = 0.94$, $P < 0.01$). (In this paper, the test results of root-soil interface characteristics under different moisture content are from Lin et al. 2024a).

Effect of moisture on the shear strength of root-soil composite

In large-scale direct shear tests, the changes in the cohesion (C_{s*}) and internal friction angle (φ_{s*}) of the R-SC with different RAR values (RAR: 0.2–1%; test numbers: SW25CR2A9–SW25CR10A9) and different soil volumetric moisture contents (θ_w : 25–45%; test numbers: SW25CR6A9–SW45CR6A9) are shown in Fig. 12. As the RAR increased, the additional strength provided by the roots increased, leading to a significant increase in C_{s*} and φ_{s*} of the R-SC (the C_{s*} increased from 33.0 kPa to 50.6 kPa, and the φ_{s*} increased from 34.47° to 39.60°). As the soil moisture content increased, the C_{s*} and φ_{s*} of the R-SC significantly decreased (C_{s*} decreased from 44.0 kPa to 13.2 kPa, and the φ_{s*} decreased from 35.81° to 30.90°). This trend was similar to the changes in the R–SI friction at different soil moisture contents described in Sect. "Effect of moisture on friction characteristics of root-soil interface", and the shear

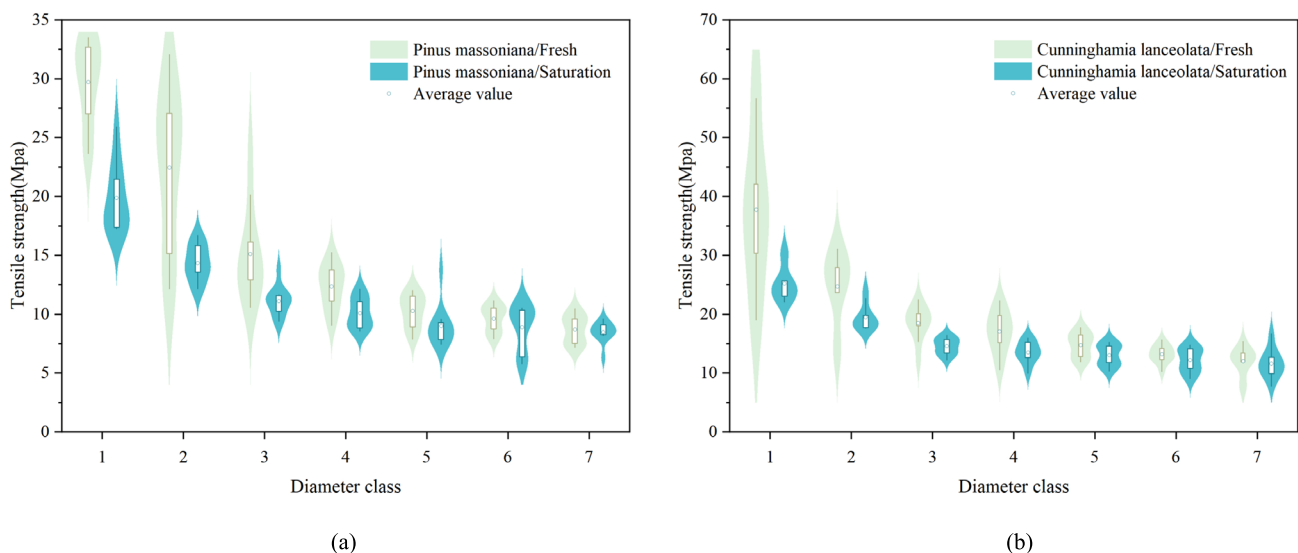


Fig. 10 Violin plots of roots tensile strength in fresh/saturated state across different diameter classes; (a) for *P. massoniana* roots, and (b) for *C. lanceolata* roots

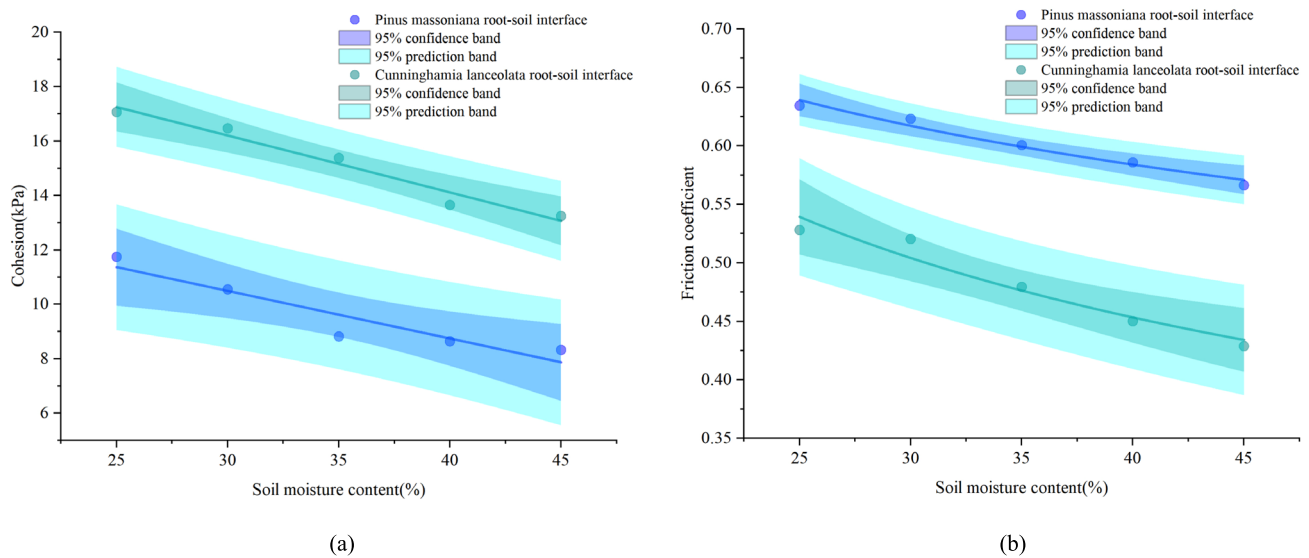


Fig. 11 Friction characteristics at the root-soil interface under different soil moisture contents (Lin et al. 2024a). (a) C_{rs} variation trend at 25–45% soil moisture content; and (b) f_{rs} variation trend at 25–45% soil moisture content

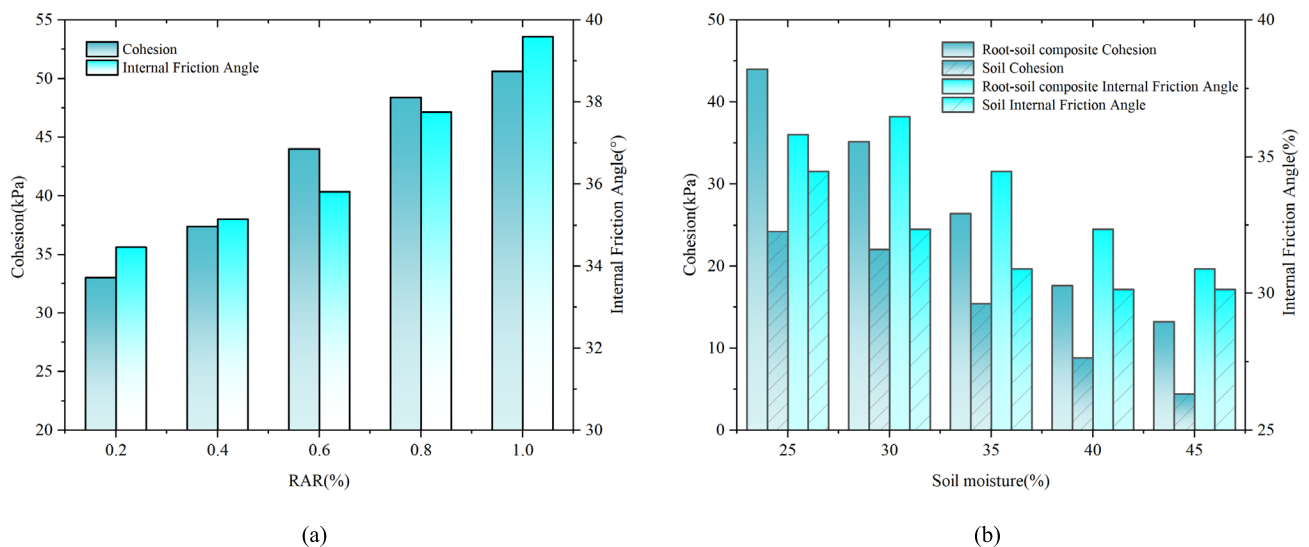


Fig. 12 Strength characteristics of the R-SC of trees in different (a) RAR values and (b) soil moisture contents

strength of the soil without roots also decreased as moisture content increased. Therefore, the deterioration of the strength of R-SC under moisture infiltration occurred due to multiple interacting factors, which were considered in subsequent calculations.

The effects of different RARs on RE_p at a 25% volumetric moisture content are shown in Fig. 13(a). With increasing RAR, RE_p increased significantly. When the RAR increased from 0.2% to 1.0%, RE_p increased from 0.14 to 0.56, showing a linear increase: $RE_p = 0.03 + 0.54\text{RAR}$ ($R^2 = 0.99$, $P < 0.01$). In regions with higher root distribution density, the soil strength increased substantially. The effects

of different soil moisture contents on RE_p at a 0.6% RAR are shown in Fig. 13(b). As the moisture content increased, RE_p decreased significantly. As θ_w rose from 25 to 45%, RE_p decreased from 0.36 to 0.29, showing a linear decrease: $RE_p = 0.45 - 0.004\theta_w$ ($R^2 = 0.94$, $P < 0.01$). After intense rainfall, the soil moisture content in shallow landslides increased significantly, negatively impacting the strength of the R-SC.

Figure 14(a) shows the shear strength of the *C. lanceolata* R-SC (θ_w : 25%, RAR: 0.4%), which is composed of non-coarse roots and coarse roots under different vertical pressures. At the same θ_w and RAR, the R-SC composed of non-coarse roots exhibited greater shear strength under

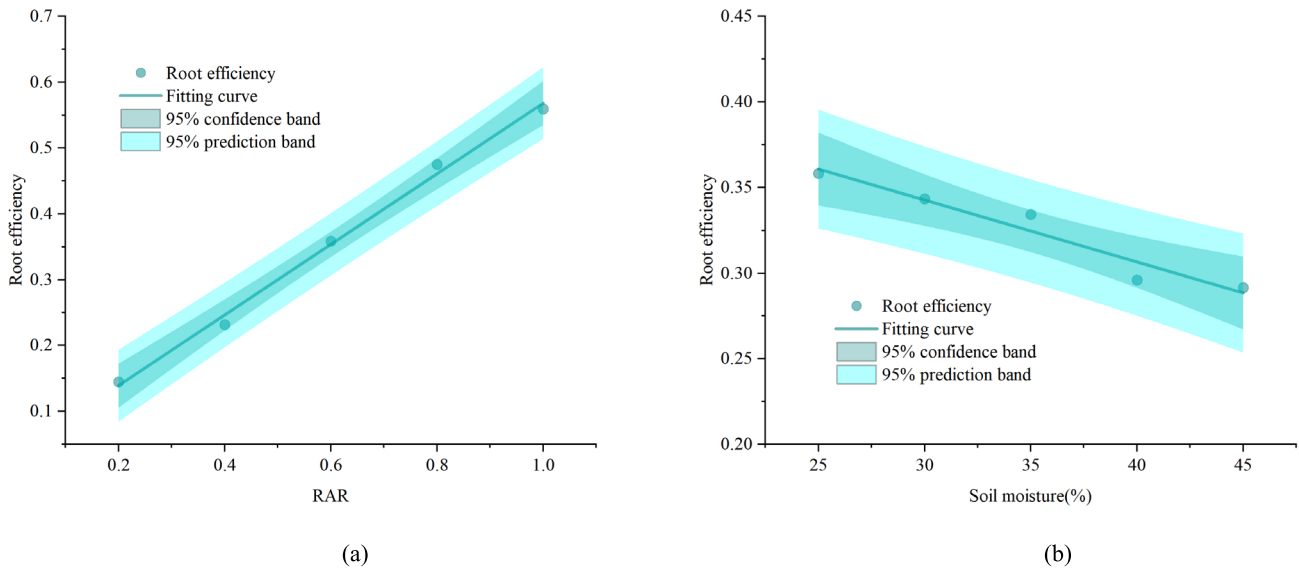


Fig. 13 Root efficiency with different (a) RAR values and (b) soil volumetric moisture contents

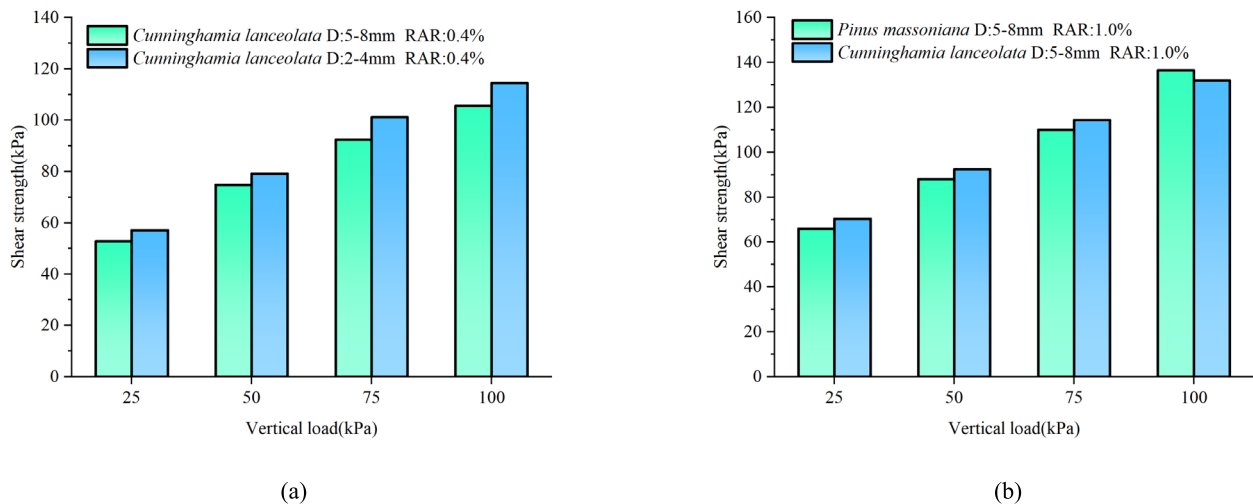


Fig. 14 Shear strength under different (a) root diameters and (b) root types

vertical pressures of 25–100 kPa, indicating an increase of about 8.02%. C_{s*} increased from 37.4 kPa to 39.6 kPa, and the φ_{s*} increased from 35.15° to 37.75°. This effect might be more pronounced when the RAR is larger. Figure 14(b) shows the shear strength of R-SC composed of *C. lanceolata* coarse roots and *P. massoniana* coarse roots (θ_w :25%,RAR: 1.0%) under different vertical pressures. Compared to *P. massoniana*, the R-SC with *C. lanceolata* coarse roots had greater shear strength under vertical pressures of 0–50 kPa. However, at a vertical pressure of 100 kPa, the *P. massoniana* R-SC had greater shear strength. The differences in the R–SI friction between the

two types of roots can explain this phenomenon. At a volumetric moisture content of 25%, the R–SI cohesion of *P. massoniana* is lower than that of *C. lanceolata*, but the R–SI friction coefficient of *P. massoniana* is higher. When the vertical pressure is low, the interface frictional resistance cannot be fully utilized, resulting in the R–SI shear strength (bonding strength) of *P. massoniana* being lower than that of *C. lanceolata*. Conversely, when the vertical pressure is high, the interface frictional resistance is fully mobilized, leading to the R–SI shear strength of *P. massoniana* exceeding that of *C. lanceolata*. This difference influences the shear strength of the R-SC.

Model validation of RSFM model

Equation (31) is fitted to the measured SWCC curve ($R^2 = 0.89$) (Lai et al. 2024) (Fig. 15):

$$\psi = \left[\left(\frac{\theta_s}{\theta} \right)^{2.96} - 1 \right] / 0.26 \quad (34)$$

For the relationship between the suction strength τ_{us} and the suction ψ of unsaturated residual sandy clay soil, fitting according to Equation (27) yield ($R^2 = 0.93$):

$$\tau_{us} = \psi \left(\frac{\psi}{\psi_a} \right)^{0.93} \tan \varphi' \quad (35)$$

The fitting of this curve was inadequate because this study focused on the strength variations of unsaturated residual sandy clay soil within the specific volumetric moisture content range of 25–45%. Equation (27) may be more suitable for other soil types or not appropriate for fitting the relationship between strength and suction within this particular moisture content range of residual sandy clay soil. In this study, Equation (36) better fits the relationship between strength and suction within the volumetric moisture content range of 25–45% in residual sandy clay soil ($R^2 = 0.93$).

$$\tau_{us} = \psi \tan 59.39^\circ \quad (36)$$

Substituting the root tensile strength and R–SI friction characteristic parameters obtained from Sects. "Influence of moisture on root tensile strength" and "Effect of moisture on friction characteristics of root-soil interface" into Eqs. (33) and (36) yielded the computed values of the RSFM model. Comparing the computed values of the RSFM model with the measured values and the computed values from the

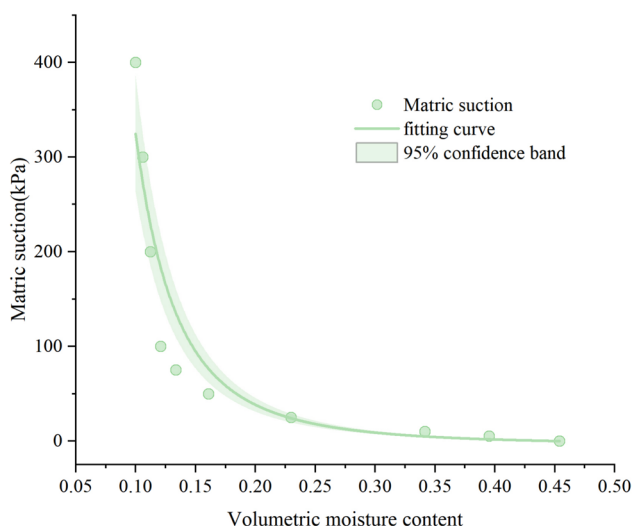


Fig. 15 SWCC curve of residual sandy clay soil

Correction Wu model (the original Wu model is overestimated on root reinforcement cohesion, a correction factor $k'' = 0.4$ was often used to reduce this kind of overestimation (Preti and Schwarz 2006; Dai et al. 2022; Jiang et al. 2024)), as shown in Fig. 16, the deviation rate (DR) of the computed model can be defined as:

$$DR = \frac{MCV - TV}{TV} \quad (37)$$

Here, DR represents the deviation rate, MCV represents the model's computed values, and TV represents the tested values.

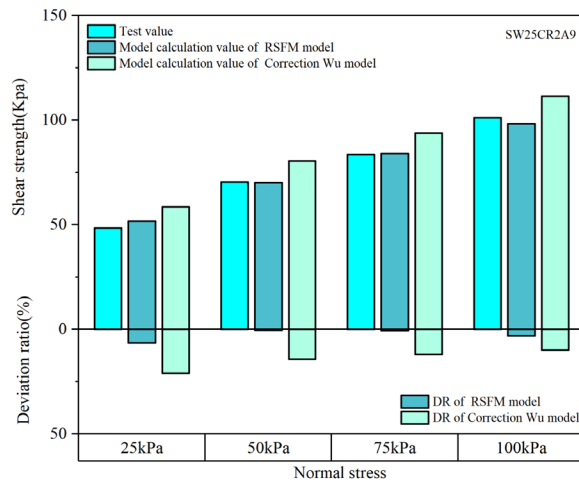
The maximum deviation rate of the RSFM model did not exceed 11% (Fig. 16), which was more accurate than Correction Wu model (average deviation rate: 50.72%) in most cases. Additionally, the deviation rates of the Correction Wu model were greater when the normal stress was lower or the soil moisture content was greater. Such a pattern occurred because lower normal stress and a higher soil moisture content decreased the effectiveness of frictional resistance at the R–SI, leading to a failure mode predominantly characterized by root pullout. In such cases, the Correction Wu model, which estimates the additional strength provided by roots based only on root tensile strength, overestimates this strength. In contrast, the RSFM model, which simultaneously considers both the R–SI friction and the root tensile strength, can reduce this error.

Discussion

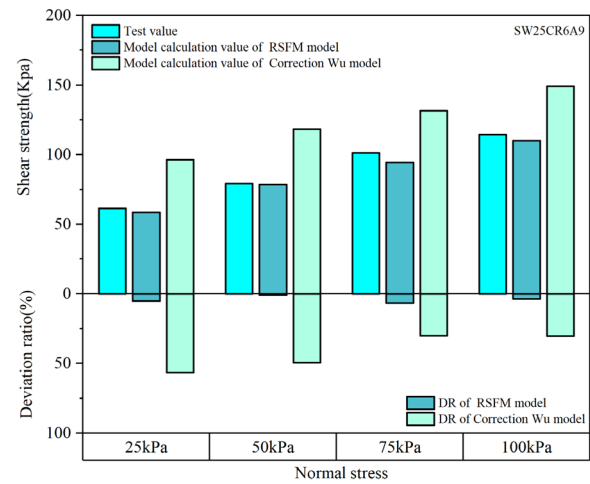
Degradation mechanism of the shear strength of root–soil composites under moisture infiltration

For R–SC, moisture infiltration affects the roots, soil, and R–SI. According to the results described in Sect. "Influence of moisture on root tensile strength", the tensile strength of tree roots decreased (especially for finer roots) when the moisture content of the roots increased from natural conditions to saturation. This result matched the findings of Zhang et al. (2019) for herbaceous plant roots and Hales et al. (2013) for woody plant roots. Moisture content significantly impacts root tensile strength. As the moisture content in roots increases, water accumulates in the cell wall, which decreases the binding strength between organic polymers in the cell wall. Since the cell wall determines the mechanical strength of plant roots (Hales and Miniati 2017; Zhang et al. 2019), testing roots under inappropriate moisture conditions for rainfall-induced vegetative slope failure may overestimate their potential to reinforce the soil under critical failure conditions.

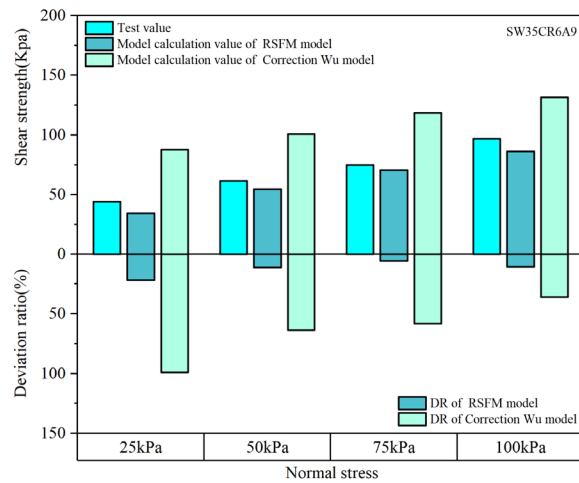
The friction characteristics of the R–SI significantly affects R–SC shear strength. Increased frictional resistance



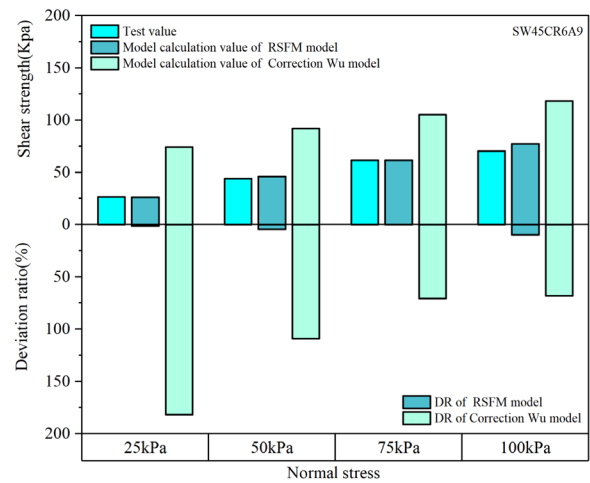
(a)



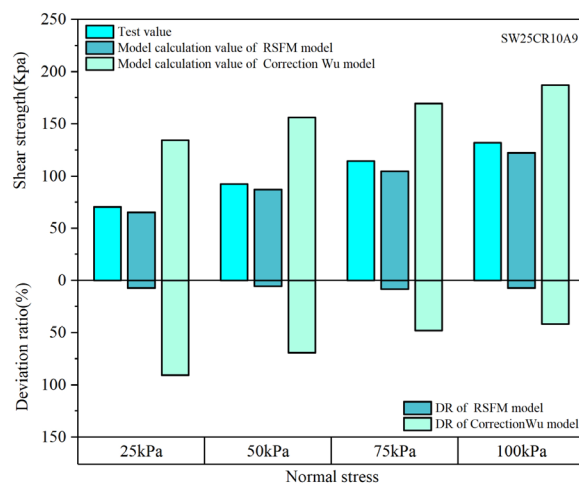
(b)



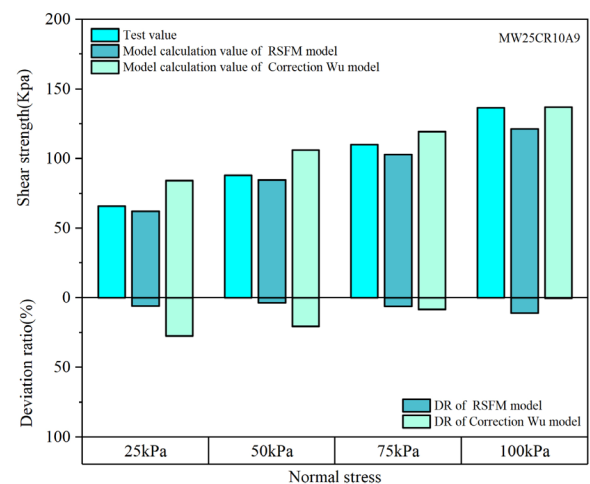
(c)



(d)



(e)



(f)

Fig. 16 Measured values and model-computed values. (a) Measured values and model-computed values for the SW25CR2A9 test.;(b) SW25CR6A9 test.;(c) SW35CR6A9 test.;(d) SW45CR6A9 test.;(e) SW25CR10A9 test.;(f) MW25CR10A9 test

between the roots and soil is vital for limiting soil deformation and improving the shear strength of the R-SC (Zhang et al. 2014). The results presented in Sect. "Effect of moisture on friction characteristics of root-soil interface" revealed that moisture significantly affects the R-SI of two types of tree roots within a volumetric moisture content range of 25–45%. Moisture influences the interfacial suction formed by pore water between roots and soil particles at the R-SI, disrupts the cementing materials at the interface, and increases the distance between soil particles and roots, thus altering the bonding force (cohesion and friction at the root-soil interface) at the R-SI (Jemberu et al. 2017; Lin et al. 2024a). For the two types of tree R-SI, the C_{rs} and f_{rs} significantly decreased with an increase in θ_w , leading to more root pull-out failures during deformation. Thus, the tensile strength of the roots cannot be fully utilized, and the additional strength provided by the roots becomes shear strength at the R-SI, resulting in a decrease in RE_p (Fig. 13(b)).

For unsaturated soils, when the moisture content increases to a certain level, the matric suction of soil significantly decreases, which affects the strength of the soil (Bishop 1959; Fredlund et al. 1978; Khalili and Khabbaz 1998). Consequently, in R-SC, as moisture infiltrates, the tensile strength of the roots, soil shear strength, and R-SI shear strength decrease to different degrees. This leads to a decrease in the ability of the R-SC to resist deformation, resulting in a decrease in its strength. Furthermore, when the moisture content of the R-SC approaches saturation, its strength is at its lowest.

Limitations of root reinforcement in vegetation-covered landslides under heavy rainfall

Although many researchers have confirmed the reinforcing effect of plant root systems on soil (Lotfalian et al. 2019; Boldrin et al. 2021) and observed significant increases in soil strength within root-dense layers, this reinforcing effect highly depends on the distribution of the root system. Many researchers have found that the sliding surface depth of some shallow landslides with vegetation cover is often near the maximum root distribution depth (Roering et al. 2003; Guo et al. 2020; Lin et al. 2024b). During rainfall, the depth of the vegetation root system strongly influences the erosion intensity of landslides. Moreover, the maximum distribution depth of plant root systems is quite limited. In some landslide-prone areas, the maximum root depth of trees is often within 2 m (Qin et al. 2022; Lin et al. 2024b), whereas shrubs and herbaceous plants have even shallower roots. Thus, the range of root reinforcement in the soil on slopes is limited. As soil depth increases, the RAR and root number density (RND) significantly decrease, leading to a substantial reduction in the additional strength provided. During

heavy rainfall, the strength of the R-SC decreases, further weakening the reinforcement effect of the roots. These factors limit the reinforcing effect of roots on slopes. Additionally, some root systems promote infiltration (Ghestem et al. 2011; Ni et al. 2019), which, under heavy rainfall, may further accelerate the decrease in strength of the R-SC and the formation of the sliding surface (Fig. 17).

Limitations of the model proposed in this study

In the failure process of the root-soil composite, roots may undergo either fail in breakage or fail in pullout. The RSFM model presented in this study accounts for both failure modes. However, for root pullout failure, there are two distinct types of pullout failure mode. The failure surface can occur not only at the root-soil interface but also within the soil mass, meaning that the root is pulled out from surrounding soils with a layer of soils covered on its root surface. This phenomenon has been reported by other researchers (Dupuy et al. 2007; Huang et al. 2024). Moreover, the pull-out failure surface within the soil mass may be irregular, making it challenging to determine the failure surface area. As a result, the additional strength provided by the root in this failure mode is more difficult to quantify. The RSFM model in this study does not account for this scenario, and future research should address this aspect.

Conclusions

To accurately assess the reinforcement effect of roots on landslides during rainfall, the mechanism underlying the degradation of shear strength in root-soil composites (R-SC) due to moisture was investigated. The findings are as follows:

For both *P. massoniana* and *C. lanceolata* roots, moisture significantly affected the tensile strength of smaller-diameter roots (diameter classes 1–3), and the tensile strength of smaller-diameter roots under saturated conditions was significantly lower than that under fresh conditions. However, for larger-diameter roots (diameter classes 6–7), moisture had a negligible effect.

As the RAR increases, the additional strength provided by the roots also increases, and the root efficiency significantly improves. When RAR increased from 0.2% to 1.0%, RE_p value increased from 0.14 to 0.56, showing a linear increase. As the moisture content increases, the additional strength provided by the roots decreases, leading to a significant reduction in root efficiency. When the volumetric soil moisture content increased from 25 to 45%, the RE_p value decreased from 0.36 to 0.29, showing a linear decrease.

For the same moisture content and RAR, we found differences in the shear strength of the R-SC depending on root

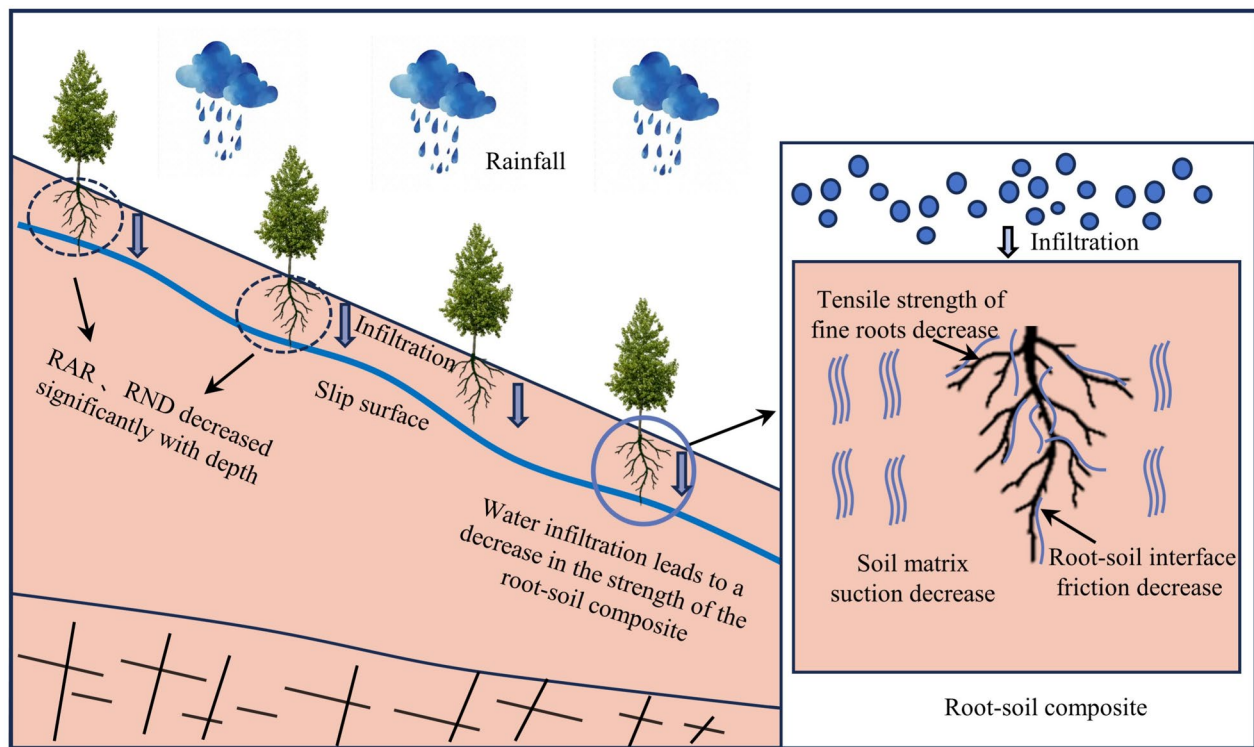


Fig. 17 Changes in root reinforcement effects under rainfall

diameter (coarse roots vs. non-coarse roots) and root type (*C. lanceolata* roots vs. *P. massoniana* roots). Under the vertical pressure of 25–100 kPa, the *C. lanceolata* R-SC composed of non-coarse roots exhibited greater shear strength than the composite composed of coarse roots, with an improvement of about 8.02%. The shear strength of the R-SC with coarse *C. lanceolata* roots was greater than that of *P. massoniana* under vertical pressures ranging from 0–50 kPa. However, at a vertical pressure of 100 kPa, the *P. massoniana* R-SC presented greater shear strength. This phenomenon can be explained by the differences in the frictional characteristics of the R-SI between the two root types.

In R-SC, as moisture infiltrates, the tensile strength of the roots, soil shear strength, and R-SI shear strength decrease to different degrees. This results in reduced resistance to deformation in the R-SC, leading to a decrease in its strength. Based on the widely applied Wu model, a degradation model for the strength of R-SC considering moisture effects was developed. The model incorporates root failure modes and the frictional mechanics of the root-soil interface. By quantifying the shear strength of the root-soil interface through its frictional characteristics, the model accounts for factors such as root diameter, root length, moisture effects on root-soil interface friction properties, and soil matric suction. The model was validated using data from the root–tensile tests, R-SI friction tests, and R-SC large-scale direct

shear tests, showing that the model's calculation results are relatively accurate.

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Data availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

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